

Answer Key — Biology Grade 11

Sub-Strand 1.2: Ecology • Lesson 8: Nitrogen and Carbon Cycles: How Nutrients Move and Why Algae Boom in Our River

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
B	C	D	D	D	C	C	D	B	C

USE AFTER: OBSERVE PHASE — Guided discussion: 'Why do farmers in Kisii and Bungoma plant beans between maize rows?'

Q1. Farmers in Kisii and Bungoma often plant beans between rows of maize. Which process in the bean roots helps the maize crop get more nitrogen?

- A. Bean roots take in nitrogen gas and release it directly to the maize roots
- B. Rhizobium bacteria in root nodules change nitrogen gas from the air into nitrogen compounds**
- C. Denitrifying bacteria in the soil change nitrates into nitrogen gas for the maize
- D. Bean leaves use photosynthesis to make nitrates for the maize

Correct answer: B. Rhizobium bacteria in bean and groundnut root nodules fix nitrogen gas into compounds that plants can use, and the soil is enriched with them. Denitrification removes usable nitrogen, so it does not help.

USE AFTER: OBSERVE PHASE — Cycle cards: nitrogen fixation, nitrification, nitrates absorbed by maize and sukuma wiki

Q2. A student says, 'Maize plants take nitrogen gas straight from the air and use it to make proteins.' Which is the best correction?

- A. Maize can only get nitrogen by eating dead animals
- B. Maize gets nitrogen from carbon dioxide during photosynthesis
- C. Maize absorbs nitrates from the soil, which were made by nitrogen fixation and nitrification**
- D. Maize absorbs nitrogen gas through its stomata, but only at night

Correct answer: C. Plants cannot use nitrogen gas directly. They absorb nitrates through their roots, and these come from fixation (bacteria, lightning) and nitrification of ammonium.

USE AFTER: OBSERVE PHASE — Circulating question: 'What happens to the nitrogen when the cow dies?'

Q3. A dairy cow in Kericho dies and its body decays in the soil. Which organisms release the nitrogen from its proteins as ammonium so that it can re-enter the nitrogen cycle?

- A. Producers such as maize
- B. Nitrogen-fixing bacteria in bean roots only
- C. Consumers such as other cattle
- D. Decomposers such as bacteria and fungi**

Correct answer: D. Decomposers break down dead organisms and release nitrogen as ammonium, which nitrifying bacteria then change to nitrates. Producers and consumers do not do this breakdown.

USE AFTER: OBSERVE PHASE — Guided discussion: 'Where does the carbon in your ugali end up when you run?'

Q4. A learner eats ugali and then runs. Which process returns carbon from the ugali to the air as carbon dioxide?

- A. Denitrification in the learner's muscles
- B. Nitrification in the learner's stomach
- C. Photosynthesis in the learner's cells
- D. Respiration in the body's cells**

Correct answer: D. Cells use respiration to release energy from food, and carbon dioxide is given off and breathed out. Photosynthesis takes CO₂ in and happens in plants, not animals.

USE AFTER: OBSERVE PHASE — Cycle cards for carbon: photosynthesis, respiration, decomposition, combustion of firewood and fossil fuels

Q5. Which list contains only processes that return carbon dioxide to the air in the carbon cycle?

- A. Photosynthesis, burning charcoal and nitrification
- B. Nitrogen fixation, respiration and burning firewood
- C. Photosynthesis, respiration and decomposition
- D. Respiration, decomposition and burning firewood**

Correct answer: D. Respiration, decomposition (microbial respiration) and combustion of firewood or fossil fuels all release CO₂. Photosynthesis removes CO₂ from the air.

USE AFTER: EXPLAIN PHASE — Groups' 4-5 sentence explanation: how too much nitrates and phosphates lead to more algae

Q6. Rain washes fertiliser from maize farms, cattle waste and sewage into a river. Why does this make algae and water weeds grow very fast?

- A. The fertiliser makes the water colder, so the algae grow faster
- B. The sewage gives the algae more oxygen to breathe
- C. The water gains extra nitrates and phosphates that the algae and weeds use to grow**
- D. The algae feed directly on the fish in the river

Correct answer: C. Fertiliser, manure and sewage add nitrates and phosphates (nutrient enrichment), which allow algae and weeds to multiply quickly.

USE AFTER: EXPLAIN PHASE — Chain scaffold: 'If decomposers use oxygen, what happens to fish that need oxygen?'

Q7. In a river with an algal bloom, the algae and weeds die in large numbers. Why do fish then die?

- A. The fish die of old age because the water is warmer
- B. The fish have no food because algae are the only food in rivers
- C. Decomposing bacteria multiply and use up dissolved oxygen, so the fish cannot get enough**
- D. The dead algae release extra oxygen that is poisonous to fish

Correct answer: C. Dead algae and weeds feed decomposers, which use dissolved oxygen in respiration. Low oxygen causes fish to suffocate, which corrects the idea that fish simply have no food.

USE AFTER: EXPLAIN PHASE — Groups sequence the chain: runoff, nutrient enrichment, algal bloom, ... fish suffocate

Q8. Which is the correct order of events in the river after fertiliser and sewage runoff enter it?

- A. Decomposers use up oxygen, nutrient enrichment, algae and weeds die, algal bloom, fish die
- B. Algal bloom, nutrient enrichment, fish die, decomposers use up oxygen, algae and weeds die
- C. Fish die, decomposers use up oxygen, nutrient enrichment, algal bloom, algae and weeds die
- D. Nutrient enrichment, algal bloom, algae and weeds die, decomposers use up oxygen, fish die**

Correct answer: D. Enrichment comes first and causes the bloom. The bloom dies and is decomposed, oxygen is used up, and fish suffocate.

USE AFTER: EXPLAIN PHASE — Chain: anaerobic decay produces foul smell (explanation writing)

Q9. Along a polluted river, people notice a foul smell after many algae and weeds have died. What is the best explanation?

- A. The smell comes from nitrogen gas escaping from the water
- B. With little oxygen left, anaerobic decay of the dead matter produces foul-smelling substances**
- C. Fish release a bad smell when they are hungry
- D. Algae produce a bad smell only when they are alive and healthy

Correct answer: B. When oxygen is used up, decay continues without oxygen (anaerobic decay) and produces foul-smelling products.

USE AFTER: END OF LESSON — Peer review and revision: noting learners who still say 'fish have no food' without mentioning oxygen

Q10. A student hears a group claim 'the fish are disappearing because they have no food.' What evidence from the river would best show this claim is incomplete?

- A. The river still has some fish in it
- B. The river is flowing towards Lake Victoria
- C. Algae and weeds are growing fast while dissolved oxygen falls after they die and decompose**
- D. Beans are planted near the river

Correct answer: C. Plenty of algae and weeds means food is not the only problem. Oxygen loss from decomposition explains the fish loss, which is the point to note when learners say fish have no food.